

Solutions of assignments for Mathematical statistics

NPU

1 Chapter 1

Omitted.

2 Chapter 2

2.2.16

If A and B are mutually exclusive events, $P(A) = 0.17$ and $P(B) = 0.46$, find

- (a) $P(A \cup B)$
- (b) $P(A^c)$
- (c) $P(A^c \cup B^c)$
- (d) $P(A \cap (B)^c)$
- (e) $P(A^c \cap B^c)$

Solution If A and B are mutually exclusive events, then

$$A \cap B = \phi, P(A \cap B) = 0$$

(a)

$$P(A \cup B) = P(A) + P(B) - P(AB) = 0.17 + 0.46 - 0 = 0.63$$

(b)

$$P(A^c) = 1 - P(A) = 1 - 0.17 = 0.83$$

(c)

$$P(A^c \cup B^c) = 1 - P(A \cap B) = 1 - 0 = 1$$

(d)

$$P(A \cap (B)^c) = P(A) - P(AB) = 0.17 - 0 = 0.17$$

(e)

$$P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - 0.63 = 0.37$$

2.2.21 A box contains three red and two blue flies. Two flies are removed with replacement. Let A be the event that both the flies are of the same color and B be the event that at least one of the flies is red. Find

- (1) $P(A)$
- (2) $P(B)$
- (3) $P(A \cup B)$
- (4) $P(A \cap B)$

Solution

(1) The sample space consists of four outcomes, namely $S = \{(red, red), (red, blue), (blue, red), (blue, blue)\}$.

Then the outcomes in A are $\{(red, red), (blue, blue)\}$. Therefore

$$P(A) = \frac{3}{5} \times \frac{3}{5} + \frac{2}{5} \times \frac{2}{5} = \frac{13}{25}.$$

(2) B^c be the event that both the flies are blue. The outcome in B^c is $\{(blue, blue)\}$. Therefore,

$$P(B) = 1 - P(B^c) = 1 - \frac{2}{5} \times \frac{2}{5} = \frac{21}{25}.$$

(3) The outcomes in $A \cup B$ are S . Therefore,

$$P(A \cup B) = 1.$$

(4) The outcome in $A \cap B$ is $\{(red, red)\}$. Therefore,

$$P(A \cap B) = \frac{3}{5} \times \frac{3}{5} = \frac{9}{25}.$$

2.3.12 A package of 15 apples contains two defective apples. Four apples are selected at random.

- (a) Find the probability that none of the selected apples is defective.
- (b) Find the probability that at least one of the selected apples is defective.

Solution (a) The number of ways of selecting four apples is

$$\binom{15}{4} = \frac{15 \times 14 \times 13 \times 12}{4 \times 3 \times 2 \times 1}$$

The number of ways of selecting none of defective apples is

$$\binom{13}{4} = \frac{13 \times 12 \times 11 \times 10}{4 \times 3 \times 2 \times 1}$$

Hence, the probability that none of the selected apples is defective is

$$\frac{\binom{13}{4}}{\binom{15}{4}} = \frac{11}{21}$$

(b) The probability that at least one of the selected apples is defective is

$$1 - \frac{\binom{13}{4}}{\binom{15}{4}} = 1 - \frac{11}{21} = \frac{10}{21}.$$

2.3.16 A box of manufactured items contains 12 items, of which four are defective. If three items are drawn at random without replacement, what is the probability that

- (a) The first one is defective and the rest are good?
- (b) Exactly one of the three is defective?

Solution

(a) Let "The first time a defective manufactured item is drawn" is A.
 Let "The remaining two times, good items are drawn." is B.
 The probability that the first one is defective and the rest are good is

$$\begin{aligned} P(A \cap B) &= P(A) * P(B | A) \\ &= \frac{4}{12} \cdot \frac{C_8^2}{C_{11}^2} \\ &= \frac{4}{12} \times \frac{8}{11} \times \frac{7}{10} \\ &= \frac{28}{165} \end{aligned}$$

(b) Let this event be C.

The probability that the second one is defective and the rest are good is

$$\frac{8}{12} \times \frac{4}{11} \times \frac{7}{10} = \frac{28}{165}.$$

The probability that the third one is defective and the rest are good is

$$\frac{8}{12} \times \frac{7}{11} \times \frac{4}{10} = \frac{28}{165}.$$

Hence, the probability that exactly one of the three is defective is

$$P(C) = \frac{28}{165} + \frac{28}{165} + \frac{28}{165} = \frac{28}{55}.$$

or

$$\begin{aligned} P(C) &= \frac{C_8^2 * C_4^1}{C_{12}^3} \\ &= \frac{28}{55} \end{aligned}$$

2.4.12 A survey asked a group of 400 people whether or not they were doing daily exercise. The responses by sex and physical activity are as in Table 1.

A person is randomly selected.

Table 1: Physical Activity Survey Results by Gender

	Male	Female
Daily exercise	50	61
No daily exercise	177	112

- (a) What is the probability that this person is doing daily exercise?
- (b) What is the probability that this person is doing daily exercise if we know that this person is a male?

Solution

(a) Let A be the event that this person is doing daily exercise. Therefore,

$$P(A) = \frac{50 + 61}{400} = \frac{111}{400}$$

(b) Let B be the event that this person is a male. Therefore,

$$P(B) = \frac{50 + 177}{400} = \frac{227}{400}$$

and

$$P(AB) = \frac{50}{400} = \frac{1}{8}.$$

Hence, the probability that this person is doing daily exercise if we know that this person is a male is

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{50}{227}.$$

2.4.23 A store has light bulbs from two suppliers, 1 and 2. The chance of supplier 1 delivering defective bulbs is 10%, whereas supplier 2 has a defective rate of 3%. Suppose 60% of the current supply of light bulbs came from supplier 1. If one of these bulbs is taken from the current supply and observed to be defective, find the probability that it came from supplier 2.

Solution

Let A_1 be the event that the bulb came from supplier 1, and A_2 be the event that the bulb came from supplier 2. Here the sample space $S = A_1 \cup A_2$. Let B be the event that the bulb is defective. We are given $P(A_1) = 0.6$, $P(A_2) = 0.4$, $P(B|A_1) = 0.1$, and $P(B|A_2) = 0.03$. Then by the total probability rule, the probability that the bulb taken from the current supply is defective is

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) = 0.6 \times 0.1 + 0.4 \times 0.03 = 0.072.$$

Now by Bayes rule, the probability that the defective bulb came from supplier 2 is

$$P(A_2|B) = \frac{P(A_2)P(B|A_2)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2)} = \frac{0.4 \times 0.03}{0.072} = \frac{1}{6}.$$

2.5.8 Suppose that Y is a continuous random variable whose pdf is given by

$$f(y) = \begin{cases} K(4y - 2y^2), & 0 < y < 2 \\ 0, & \text{elsewhere} \end{cases}$$

- (a) What is the value of K ?
- (b) Find $P(Y > 1)$.
- (c) Find $F(y)$.

Solution

- (a) For $f(y)$ to be a pdf, we need $\int_{-\infty}^{\infty} f(y) dy = 1$
Therefore,

$$1 = \int_{-\infty}^{\infty} f(y) dy = \int_0^2 K(4y - 2y^2) dy = K \left(2y^2 - \frac{2}{3}y^3 \right) \Big|_0^2 = \frac{8}{3}K.$$

Hence,

$$K = \frac{3}{8}.$$

- (b) The probability of $Y > 1$ is

$$P(Y > 1) = \int_1^{\infty} f(y) dy = \int_1^2 \frac{3}{8}(4y - 2y^2) dy = \frac{3}{8} \left(2y^2 - \frac{2}{3}y^3 \right) \Big|_1^2 = \frac{1}{2}.$$

- (c) The cumulative distribution function is

$$F(y) = \begin{cases} 0, & y < 0 \\ \frac{3}{4}y^2 - \frac{1}{4}y^3, & 0 \leq y < 2 \\ 1, & y \geq 2 \end{cases}$$

2.6.6 Let X be a random variable with pdf $f(x) = kx^2$ where $0 \leq x \leq 1$.

- (a) Find k .
- (b) Find $E(X)$ and $Var(X)$.
- (c) Find $M_X(t)$. Using the mgf, find $E(X)$.

Solution

(a) For $f(x)$ to be a pdf, we need $\int_{-\infty}^{\infty} f(x) dx = 1$.

Therefore,

$$1 = \int_{-\infty}^{\infty} f(x) dx = \int_0^1 kx^2 dx = k \left(\frac{1}{3}x^3 \right) \Big|_0^1 = \frac{1}{3}k.$$

Hence,

$$k = 3.$$

(b) By definition, the mathematical expectation of X is

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx = \int_0^1 x \cdot 3x^2 dx = \frac{3}{4}x^4 \Big|_0^1 = \frac{3}{4}$$

and the mathematical expectation of X^2 is

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^1 x^2 \cdot 3x^2 dx = \frac{3}{5}x^5 \Big|_0^1 = \frac{3}{5}.$$

Therefore, the variance of X is

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{3}{5} - \left(\frac{3}{4} \right)^2 = \frac{3}{80}.$$

(c) By definition, the mgf of X is

$$M_X(t) = E(e^{tX}) = \int_0^1 e^{tx} \cdot 3x^2 dx = \begin{cases} \left(3x^2 \cdot \frac{1}{t} e^{tx} - 6x \cdot \frac{1}{t^2} e^{tx} + 6 \cdot \frac{1}{t^3} e^{tx} \right) \Big|_0^1 = 3e^t \left(\frac{1}{t} - \frac{2}{t^2} + \frac{2}{t^3} \right) - \frac{6}{t^3}, & t > 0 \\ \int_0^1 3x^2 dx = x^3 \Big|_0^1, & t = 0 \end{cases}$$

Taking the derivative of $M_X(t)$ with respect to $t = 0$, we obtain

$$\begin{aligned}
M'_X(0) &= \lim_{t \rightarrow 0^+} \frac{M_X(t) - M_X(0)}{t - 0} \\
&= \lim_{t \rightarrow 0^+} \frac{3e^t \left(\frac{1}{t} - \frac{2}{t^2} + \frac{2}{t^3} \right) - \frac{6}{t^3} - 1}{t} \\
&= \lim_{t \rightarrow 0^+} \frac{3e^t(t^2 - 2t + 2) - 6 - t^3}{t^4} \\
&= \lim_{t \rightarrow 0^+} \frac{3e^t - 3}{4t} \\
&= \lim_{t \rightarrow 0^+} \frac{3e^t}{4} \\
&= \frac{3}{4}
\end{aligned}$$

Therefore, the mathematical expectation of X is

$$E(X) = M'_X(0) = \frac{3}{4}$$

2.6.8 Given that X_1, X_2, X_3 and X_4 are independent random variables with mean 2, find $E(Y)$ and $E(Z)$ for

$$Y = 3X_4 - X_1 + \frac{1}{5}X_3$$

$$Z = X_2 + 7X_3 - 9X_1.$$

Solution

Given $E(X_1) = 2, E(X_2) = 2, E(X_3) = 2$ and $E(X_4) = 2$, the mathematical expectation of Y is

$$E(Y) = E\left(3X_4 - X_1 + \frac{1}{5}X_3\right) = 3E(X_4) - E(X_1) + \frac{1}{5}E(X_3) = 3 \times 2 - 2 + \frac{1}{5} \times 2 = \frac{22}{5}$$

and the mathematical expectation of Z is

$$E(Z) = E(X_2 + 7X_3 - 9X_1) = E(X_2) + 7E(X_3) - 9E(X_1) = 2 + 7 \times 2 - 9 \times 2 = -2.$$

2.6.16 Find $E(X)$ and $Var(X)$ for a random variable X with pdf $f(x) = \frac{1}{2}e^{-|x|}, -\infty < x < \infty$. Also find the mgf of X .

Solution

By definition, the mathematical expectation of X is

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx = \int_{-\infty}^{\infty} \frac{1}{2}xe^{-|x|} dx = 0$$

and the mathematical expectation of X^2 is

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \int_{-\infty}^{\infty} \frac{1}{2}x^2 e^{-|x|} dx = \int_0^{\infty} x^2 e^{-x} dx = (-x^2 e^{-x} - 2x e^{-x} - 2e^{-x}) \Big|_0^{\infty} = 2.$$

Therefore, the variance of X is

$$Var(X) = E(X^2) - [E(X)]^2 = 2 - 0^2 = 2.$$

By definition, the mgf of X is ($t < 1$)

$$M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{2}e^{-|x|} dx = \frac{1}{2} \int_{-\infty}^0 e^{(t+1)x} dx + \frac{1}{2} \int_0^{\infty} e^{(t-1)x} dx = \frac{1}{2} \times \frac{1}{t+1} + \frac{1}{2} \times \frac{1}{1-t} = \frac{1}{1-t^2}.$$

3 Chapter 3

3.2.5 Let $X \sim N(10, 25)$, Compute:

- (a) $P(X \leq 20)$
- (b) $P(X > 5)$
- (c) $P(12 \leq X \leq 15)$
- (d) $P(|X - 12| \leq 15)$

Solution Introduce z-transform, that is

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 10}{5}$$

(a) Using the z-transform, we have:

$$\begin{aligned} P(X \leq 20) &= P\left(Z \leq \frac{20 - 10}{5}\right) \\ &= P(Z \leq 2) \\ &\approx 0.97725. \end{aligned}$$

(b) Using the z-transform, we have:

$$\begin{aligned} P(X > 5) &= P\left(Z > \frac{5 - 10}{5}\right) \\ &= P(Z > -1) \\ &= P(Z \leq 1) \\ &\approx 0.841. \end{aligned}$$

(c) Using the z-transform, we have:

$$\begin{aligned} P(12 \leq X \leq 15) &= P\left(\frac{12 - 10}{5} \leq Z \leq \frac{15 - 10}{5}\right) \\ &= P(0.4 \leq Z \leq 1) \\ &= P(Z \leq 1) - p(Z \leq 0.4) \\ &= 0.8413 - 0.6554 \\ &\approx 0.1859. \end{aligned}$$

(d) Using the z-transform, we have:

$$\begin{aligned} P(|X - 12| \leq 15) &= P(-15 \leq X - 12 \leq 15) \\ &= P\left(\frac{-15 + 12}{5} \leq Z \leq \frac{15 + 12}{5}\right) \\ &= P(-2.6 \leq Z \leq 3.4) \\ &= P(Z \leq 3.4) - p(Z \leq -2.6) \\ &= P(Z \leq 3.4) - [1 - p(Z \leq 2.6)] \\ &\approx 0.995. \end{aligned}$$

3.2.15 The efficiency X of a certain electrical component may be assumed to be a random variable that is distributed uniformly between 0 and 100 units. What is the probability that X is:

- (a) Between 60 and 80 units?

- (b) Greater than 90 units?
 (c) Interpret (a) and (b).

Solution For a random variable X uniformly distributed over the interval $[a, b]$, the probability density function is:

$$f(x) = \frac{1}{b-a} = \frac{1}{100-0} = \frac{1}{100}, \quad 0 \leq x \leq 100$$

- (a) To find the probability over the interval $[60, 80]$:

$$P(60 \leq X \leq 80) = \int_{60}^{80} f(x) dx = \int_{60}^{80} \frac{1}{100} dx = \frac{1}{100} \times (80 - 60) = \frac{1}{100} \times 20 = 0.2$$

- (b) For $X > 90$, we have:

$$P(X > 90) = \int_{90}^{100} f(x) dx = \int_{90}^{100} \frac{1}{100} dx = \frac{1}{100} \times (100 - 90) = \frac{1}{100} \times 10 = 0.1$$

- (c) The result in (a) shows that the probability of X being between 60 and 80 is 0.2, indicating a 20% chance that the component's efficiency falls within this range.

The result in (b) shows that the probability of X being greater than 90 is 0.1, indicating a 10% chance that the component's efficiency exceeds 90.

3.2.18 Suppose that the life length of a mechanical component is normally distributed

- (a) If $\sigma = 3$ and $\mu = 100$, find the reliability of such a system at 105 h.
 (b) What should be the expected life of the component if it has reliability of 0.90 for 120 h?

Solution

- (a) The reliability of such a system at 105 h, represents the probability that the component will last longer than 105 hours. Therefore, we need to calculate $P(X > 105)$, where $X \sim N(100, 3^2)$. Introduce z-transform, that is

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 100}{3} \sim N(0, 1)$$

Therefore,

$$\begin{aligned} P(X \geq 105) &= 1 - P(X \leq 105) \\ &= 1 - P\left(Z \leq \frac{5}{3}\right) \\ &= 0.04746. \end{aligned}$$

- (b) This question asks for a new mean μ' such that the reliability at 120 hours is 0.90. In other words, we have $P(X > 120) = 0.90$ where $X \sim N(\mu', 3^2)$. Introduce z-transform, that is

$$Z = \frac{X - \mu'}{\sigma} = \frac{X - \mu'}{3} \sim N(0, 1)$$

Therefore,

$$\begin{aligned} P(X > 120) &= P\left(Z \geq \frac{120 - \mu'}{3}\right) = 0.9 \\ \frac{\mu' - 120}{3} &= 1.28 \end{aligned}$$

$$\mu' = 123.84.$$

The expected life of the component should be $\mu' \approx 123.84$ hours to ensure a reliability of 90% at 120 hours.

3.2.21 A professor in a large statistics class has a grading policy such that only the 15% of the students with the highest scores will receive the grade A. The mean score for this class is 72 with a standard deviation of 6. Assuming that all the grades for this class follow a normal probability distribution, what is the minimum score that a student in this class has to get to receive an A grade?

Solution

Denote the grades for this class as the variable X , then $X \sim N(72, 36)$.

Using the z-transform, we have $Z = \frac{X-72}{6} \sim N(0, 1)$.

$$\begin{aligned} P(X \geq x_0) &= 1 - P(X \leq x_0) \\ &= 1 - P\left(Z \leq \frac{x_0 - 72}{6}\right) \\ &= 0.15. \end{aligned}$$

Therefore,

$$P\left(Z \leq \frac{x_0 - 72}{6}\right) = 0.85$$

Using the normal table, we have

$$\begin{aligned} P(Z \leq 1.04) &\approx 0.85 \\ \frac{x_0 - 72}{6} &= 1.04 \\ x_0 &= 78.24 \end{aligned}$$

3.3.6 Find the marginal density functions of the random variables X and Y if their joint pdf is given by:

$$f(x, y) = \begin{cases} \frac{1}{5}(3x - y), & 1 \leq x \leq 2, 1 \leq y \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Solution

(1) Find the marginal density function of X , $f_X(x)$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_1^3 f(x, y) dy = \int_1^3 \frac{1}{5}(3x - y) dy = \frac{1}{5}(3xy - \frac{y^2}{2}) \Big|_1^3.$$

$$\begin{aligned} \left[3xy - \frac{y^2}{2}\right]_1^3 &= \left(3x(3) - \frac{3^2}{2}\right) - \left(3x(1) - \frac{1^2}{2}\right) \\ &= \left(9x - \frac{9}{2}\right) - \left(3x - \frac{1}{2}\right) \\ &= 9x - \frac{9}{2} - 3x + \frac{1}{2} \\ &= 6x - 4. \end{aligned}$$

Thus, we have:

$$f_X(x) = \frac{1}{5}(6x - 4) = \frac{6x - 4}{5}, \quad 1 \leq x \leq 2.$$

Outside this interval, $f_X(x) = 0$.

(2) Find the marginal density function of Y , $f_Y(y)$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_1^2 f(x, y) dx = \int_1^2 \frac{1}{5}(3x - y) dx = \frac{1}{5} \left(\frac{3x^2}{2} - yx \right) \Big|_1^2$$

$$\int (3x - y) dx = \frac{3x^2}{2} - yx.$$

Evaluating this from 1 to 2:

$$\begin{aligned} \left[\frac{3x^2}{2} - yx \right]_1^2 &= \left(\frac{3(2^2)}{2} - y(2) \right) - \left(\frac{3(1^2)}{2} - y(1) \right) \\ &= \left(\frac{12}{2} - 2y \right) - \left(\frac{3}{2} - y \right) \\ &= (6 - 2y) - \left(\frac{3}{2} - y \right) \\ &= 6 - 2y - \frac{3}{2} + y \\ &= \frac{12}{2} - \frac{3}{2} - 2y + y \\ &= \frac{9}{2} - y. \end{aligned}$$

Thus, we have:

$$f_Y(y) = \frac{1}{5} \left(\frac{9}{2} - y \right) = \frac{9 - 2y}{10}, \quad 1 \leq y \leq 3.$$

Outside this interval, $f_Y(y) = 0$.

The marginal density functions are:

$$f_X(x) = \begin{cases} \frac{6x - 4}{5}, & 1 \leq x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{9 - 2y}{10}, & 1 \leq y \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

3.3.8 Find k so that $f(x, y) = kxy, 1 \leq x \leq y \leq 2$, will be a pdf. Also find

- (i) $P(X \leq \frac{3}{2}, Y \leq \frac{3}{2})$,
- (ii) $P(X + Y \leq \frac{3}{2})$.

Solution

Because

$$\iint_R f(x, y) dx dy = 1.$$

Here, R is defined by the limits $1 \leq x \leq 2$ and $x \leq y \leq 2$. Thus, the double integral can be expressed as:

$$\int_1^2 \int_x^2 kxy dy dx = 1.$$

Because

$$\int_x^2 kxy dy = kx \int_x^2 y dy = kx \left(2 - \frac{x^2}{2} \right).$$

Thus, we have

$$\int_1^2 kx \left(2 - \frac{x^2}{2}\right) dx = k \int_1^2 \left(2x - \frac{x^3}{2}\right) dx.$$

We can compute this integral:

$$\int \left(2x - \frac{x^3}{2}\right) dx = \frac{x^2}{2} - \frac{x^4}{8} \Big|_1^2 = 1.125$$

Therefore, we have:

$$k \cdot 1.125 = 1 \implies k = \frac{1}{1.125} = \frac{8}{9}.$$

(i) Finding the probabilities

$$P\left(X \leq \frac{3}{2}, Y \leq \frac{3}{2}\right) = \int_1^{\frac{3}{2}} \int_x^{\frac{3}{2}} \frac{8}{9} xy \, dy \, dx.$$

$$\int_x^{\frac{3}{2}} \frac{8}{9} xy \, dy = \frac{8}{9} x \int_x^{\frac{3}{2}} y \, dy = \frac{8}{9} x \left(\frac{y^2}{2} \Big|_x^{\frac{3}{2}} \right) = \frac{8}{9} x \left(\frac{\left(\frac{3}{2}\right)^2}{2} - \frac{x^2}{2} \right) = x - \frac{4x^3}{9}$$

Thus,

$$P\left(X \leq \frac{3}{2}, Y \leq \frac{3}{2}\right) = \int_1^{\frac{3}{2}} \left(x - \frac{4x^3}{9}\right) dx = \frac{x^2}{2} - \frac{x^4}{9} \Big|_1^{\frac{3}{2}} = \frac{25}{144}$$

(ii) Calculate $P(X + Y \leq \frac{3}{2})$

Since X must be at least 1 and Y must also be at least 1, the minimum value of $X + Y$ is 2. Thus, it is impossible for $X + Y \leq \frac{3}{2}$.

Therefore,

$$P(X + Y \leq \frac{3}{2}) = 0.$$

3.3.9 The random variables X and Y have a joint pdf:

$$f(x, y) = \begin{cases} \frac{8}{9}xy, & 1 \leq x \leq y \leq 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find:

(a) The marginal of X .

(b) $P(1.5 < X < 1.75, Y > 1)$.

Solution

(a)

$$f_X(x) = \int_x^2 f(x, y) \, dy = \int_x^2 \frac{8}{9} xy \, dy = \frac{8x}{9} \left(\frac{y^2}{2} \Big|_x^2 \right) = \frac{4}{9} x(4 - x^2).$$

Thus, the marginal pdf of X is

$$f_X(x) = \frac{4}{9} x(4 - x^2), \quad 1 \leq x \leq 2.$$

(b)

$$P(1.5 < X < 1.75, Y > 1) = \int_{1.5}^{1.75} \int_x^2 f(x, y) \, dy \, dx = \int_{1.5}^{1.75} \frac{4}{9} x(4 - x^2) \, dx = \frac{8x^2}{9} - \frac{x^4}{9} \Big|_{1.5}^{1.75}$$

Thus,

$$P(1.5 < X < 1.75, Y > 1) = \frac{559}{2304} \approx 0.243.$$

3.4.2 The joint pdf of (X, Y) is:

$$f(x, y) = \frac{1}{\theta^2} e^{-\frac{x+y}{\theta}}, \quad x, y > 0, \theta > 0.$$

Find the pdf of $U = X - Y$.

Solution

$$u = x - y, y = x - u.$$

because $Y = X - U > 0, X > U$.

If $U > 0$

$$\begin{aligned} f_u(u) &= \int_u^{+\infty} \frac{1}{\theta^2} e^{-\frac{2x-u}{\theta}} dx \\ &= -\frac{1}{2\theta} e^{-\frac{2x-u}{\theta}} \Big|_u^{+\infty} \\ &= \frac{1}{2\theta} e^{-\frac{u}{\theta}} \end{aligned}$$

If $U < 0$

$$\begin{aligned} f_u(u) &= \int_0^{+\infty} \frac{1}{\theta^2} e^{-\frac{2x-u}{\theta}} dx \\ &= -\frac{1}{2\theta} e^{-\frac{2x-u}{\theta}} \Big|_0^{+\infty} \\ &= \frac{1}{2\theta} e^{\frac{u}{\theta}} \end{aligned}$$

Thus,

$$f_u(u) = \frac{1}{2\theta} e^{-\frac{|u|}{\theta}} - \infty < u < +\infty$$

3.5.4 Suppose that the number of cars arriving at a busy intersection in a given 20-min interval in a large city has a Poisson distribution with mean 120. Determine a lower bound for the probability that the number of cars arriving in a given 20-min period will be between 100 and 140 using Chebyshev's inequality. **Solution** Chebyshev's inequality is given by:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2},$$

where $k > 0$.

In this problem, we have $\mu = 120$, variance $\sigma^2 = 120$

$$k = \frac{20}{\sqrt{120}} = \frac{20}{10.95} \approx 1.83.$$

Substituting $k = 1.83$ into Chebyshev's inequality, we get:

$$P(|X - 120| < 20) \geq 1 - \frac{1}{(1.83)^2}.$$

Therefore,

$$P(100 \leq X \leq 140) = P(|X - 120| \leq 20) \geq 1 - 0.298 = 0.702.$$

3.5.6 How large should the size of a random sample be so that we can be 90% certain that the sample mean $\bar{X}$ will not deviate from the true mean by more than $\sigma/2$?

Solution According to the meaning of the question, we have: $E(\bar{X}) = \mu$, $Var(\bar{X}) = \frac{\sigma^2}{n}$, n is the size of a random sample.

That is,

$$P\left\{|\bar{X} - \mu| \leq \frac{\sigma}{2}\right\} \geq 90\%$$

From the CLT,

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

Therefore,

$$P\left\{\frac{|\bar{X} - \mu|}{\frac{\sigma}{\sqrt{n}}} \leq \frac{\sqrt{n}}{2}\right\} = P\left\{|Z| \leq \frac{\sqrt{n}}{2}\right\} \geq 90\%$$

Hence, from the normal table,

$$\begin{aligned}\frac{\sqrt{n}}{2} &\geq 1.645 \\ n &\geq 10.75\end{aligned}$$

The minimum value of n is 11.

3.5.13 Suppose that 2500 customers subscribe to a telephone exchange. There are 80 trunk lines available. Anyone customer has the probability of 0.03 of needing a trunk line on a given call. Consider the situation as 2500 trials with probability of success $p = 0.03$. What is the approximate probability that the 2500 customers will tie up the 80 trunk lines at any given time?

Solution Since $n = 2500$ is large and $p = 0.03$ is small, we can approximate this binomial distribution with a normal distribution by CLT. For a binomial distribution, the mean and variance are given by:

$$\mu = np = 2500 \times 0.03 = 75$$

$$\sigma^2 = np(1 - p) = 2500 \times 0.03 \times (1 - 0.03) = 72.75$$

Thus, the standard deviation is:

$$\sigma = \sqrt{72.75} \approx 8.53$$

$$P(X \geq 80) \approx P\left(Z \geq \frac{80 - 75}{8.53}\right) = P\left(Z \geq \frac{5}{8.53}\right) = P(Z \geq 0.585)$$

From the standard normal distribution table, we find:

$$P(Z \geq 0.585) \approx 0.281$$